SMART INDIA HACKATHON 2025

- **Problem Statement ID – 25053**
- **Problem Statement Title-** Improved Onion Storage Technology for Enhancing Shelf Life
- **Theme-** Sustainable IoT Hardware for Onion Storage.
- **PS Category-** Hardware
- **Team ID- 68904**
- **Team Name:** Team AgriNova

Team
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Proposed Solution

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We present the Improved Onion Storage Technology, a sustainable solution engineer to combat catastrophic storage losses. This system achieves maximum efficiency through biochar-mycelium trays for natural humidity control. Our unique spoilage detector is the critical feature, precisely identifying H2S gas to prevent widespread rot, resulting in a loss reduction of up to 70%.Economically, the system is superior. By utilizing agricultural waste for material production, this is an economical, one-time investment projected to save producers up to ₹30,000 per tonne.

How it addresses the problem:

- Creation of **optimum temperature and moisture buffer** to prevent sprouting and rotting of onions.
- Usage of **spoilage detection system** helps in removal of particular onion increasing shelf life of the rest.
- The biochar trays and mycelium panels use agro-waste from other farms forming a **symbiotic relationship between farmers** resulting in **cost efficiency**

Innovation & Uniqueness

- **Lo-Ra** is used to message about spoilage via our app. It uses **low power, wide area network** ideal for large farms with low access to energy.
- Spoilage detection system using sensors, this helps to pinpoint to a certain extent **the area of spoilage**.
- Farmers **don't need any prerequisite knowledge**; there will be an **indicator** along with crate number given to help them identify

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TECHNICAL APPROACH

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Air Column: An air gap between two layers of walls acts as a simple thermal barrier and moisture buffer.

MQ 135: Identifies the spoilage volatiles helping identify the area for removal

Mycelium walls: Heat insulation and moisture buffer, made from agro-waste

DHT 22: It is set at an optimum temperature, on detecting temp spikes it will operate the fans and inlet.

Radiative Roof: Reflects sunlight and emits heat, keeping the temp lower

Indicators: MQ135 examines the air in the container, the indicator connected to it signals the spoilage present

Air Inlet: Fans and air inlet uses sensors to detect temperature spikes, stabilizing the optimum temperature

Biochar Trays: Absorbs excess humidity and the gases which fasten the spoilage(eg:H2S and Ammonia)

Temperature Control

Spoilage Detection

WORK FLOW

Radiative roofs, mycelium walls cuts the heat load without electricity → Incase there is a temperature spikes, DHT22 detects it → After the temp is stabilized, the fans, inlets stop saving electricity → Air inlets and fans are used to regulate the temperature inside the unit

After spoilage, H2S gas is released → MQ 135 is used to detect H2S, which alerts the area of spoilage → The area of spoilage is found and removed using the sensor matrix → The information thus collected is sent to the app on the farmer's phone

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FEASIBILITY AND VIABILITY

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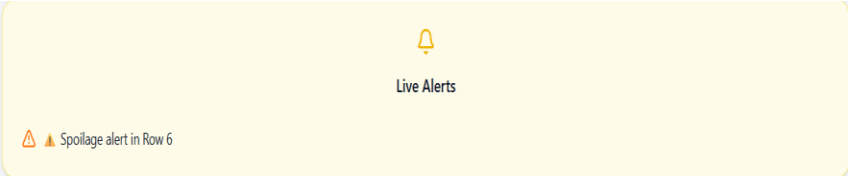
Feasibility	Challenges	Strategies to Overcome Challenges
Locally available materials (rice husk, straw, biochar).	PCM life cycle and regeneration	Regenerable biochar trays via sun-drying
Low-cost IoT hardware.	Frequent power cut in villages for prolonged period	Due to low power consumption of sensors, we can install few solar panels to tackle the problem
Simple mechanical setup.	Farmer adaptation to SMS-based alerts	Clear, concise SMS format
Modular design allows easy assembly and maintenance at farm level.	Connectivity issues in remote rural areas for SMS-based system.	Use low-power LoRa communication for better long-range coverage in rural locations.

Snippets of Our App Working



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IMPACT AND BENEFITS

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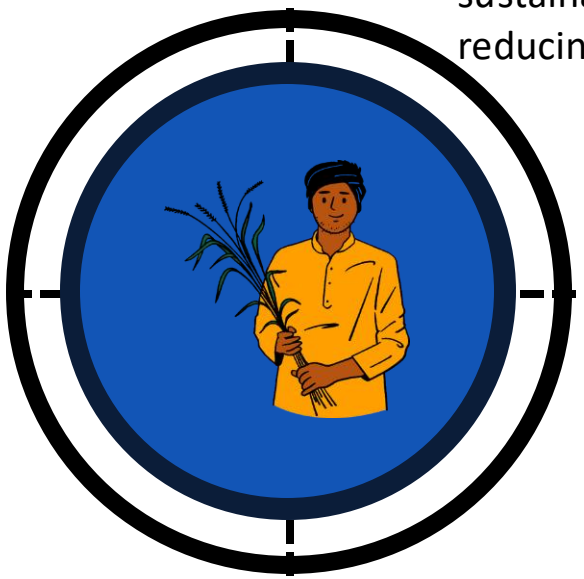
Sustainable Waste Utilization

Waste materials that are usually discarded are repurposed into valuable resources, creating sustainable solutions that directly benefit farmers by reducing costs and supporting eco-friendly practices

Minimizing Post-Harvest Loss

By reducing post-harvest loss, farmers can save their crops and increase their earnings, with each tonne saved potentially adding over ₹6,000 to their income.



Scaling Up Success:

The idea proposed is easily scalable into larger prospect with the profit being equally proportionate, this makes our idea appealing not only to small farmers, but larger ones too

Energy-Efficient Design

Optimized for minimal power consumption, enabling sensors and LoRa modules to run efficiently with extended battery life and low maintenance needs.

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RESEARCH AND REFERENCES

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- Radiative cooling research:
https://www.mdpi.com/2071-1050/15/17/13217

- Mycelium insulation studies:
https://advanced.onlinelibrary.wiley.com/doi/10.1002/adma.202409738

- PCM applications:
https://www.sciencedirect.com/science/article/pii/S2352152X25001045

- Biochar properties:
https://www.nature.com/articles/s41598-025-11580-z

References